

(odd qns first followed by even ones)

Q1)  $H$ : no. of heads,  $T$ : no. of tails

$$H - T = X, \quad H + T = n$$

$$\Rightarrow 2H = X + n \Rightarrow \boxed{X = 2H - n}$$

$H$  can take values in  $\{0, 1, 2, \dots, n\}$

$\Rightarrow X$  can take values in  $\{-n, 2-n, 4-n, \dots, n-2, n\}$

$H$ : binom( $n, p$ )

$$P_H(k) = {}^nC_k p^k (1-p)^{n-k}$$

$$\Rightarrow P_X(x) = \begin{cases} P_H(0) & , \quad x = -n \\ P_H(1) & , \quad x = 2-n \\ P_H(2) & , \quad x = 4-n \\ \vdots & \\ P_H(n) & , \quad x = n \end{cases}$$

$$= \begin{cases} {}^nC_0 (1-p)^n & , \quad x = -n \\ {}^nC_1 (1-p)^{n-1} p & , \quad x = 2-n \\ \vdots & \\ {}^nC_n (p)^n & , \quad x = n \end{cases}$$

Q3) a)  $X$ : no. of units from A to B

$$P(X=x | A \text{ woke}) = \begin{cases} p, & x = 1 \\ 1-p, & x = -\frac{3}{4} \end{cases}$$

$\nearrow$  -ve implies B given to A

$$\Rightarrow E[X | \underset{\substack{\uparrow \\ A \text{ woke}}}{A \text{ woke}}}] = p - \frac{3}{4}(1-p) = \frac{7p-3}{4}$$

$$b) P(X/A \text{ wrote } 2) = \begin{cases} 1-p, & x=2 \\ p, & x=-\frac{3}{4} \end{cases}$$

$$E[X|A \text{ wrote } 2] = 2(1-p) - \frac{3}{4}p \\ = 2 - \frac{11p}{4}$$

let. ~~P~~  $Y$ : no. what A writes down

$$P(Y=y) = \begin{cases} k, & y=1 \\ 1-k, & y=2 \end{cases}$$

$$E[X] = E[X|Y=1]P(Y=1) + E[X|Y=2]P(Y=2)$$

$$= \left(\frac{7p-3}{4}\right)k + \left(2 - \frac{11p}{4}\right)(1-k)$$

$$= \left(\frac{-18p-11}{4}\right)k + 2 - \frac{11p}{4}$$

Case I:  $-18p-11 \geq 0 \Rightarrow p \geq \frac{11}{18}$

$$\Rightarrow \min(E[X]) = 2 - \frac{11p}{4} \quad (k=0)$$

$$\Rightarrow \arg\max_p \min(E[X]) = \arg\max_p \left(2 - \frac{11p}{4}\right) \\ = \frac{11}{18}$$

Case II:  $-18p-11 < 0 \Rightarrow p < \frac{11}{18}$

$$\arg\max_p \min(E[X]) = \arg\max_p \left(\frac{18p-11}{4} + 2 - \frac{11p}{4}\right) \quad \text{when } k=1$$

$$= \arg\max_p \left(\frac{7p-3}{4}\right)$$

$$= \frac{11}{18}$$

Expected gain seen by B.

$$E[X] = \left( \frac{18p-11}{4} \right) k + \left( 2 - 11 \frac{p}{4} \right) (1-k) = f(p, k)$$

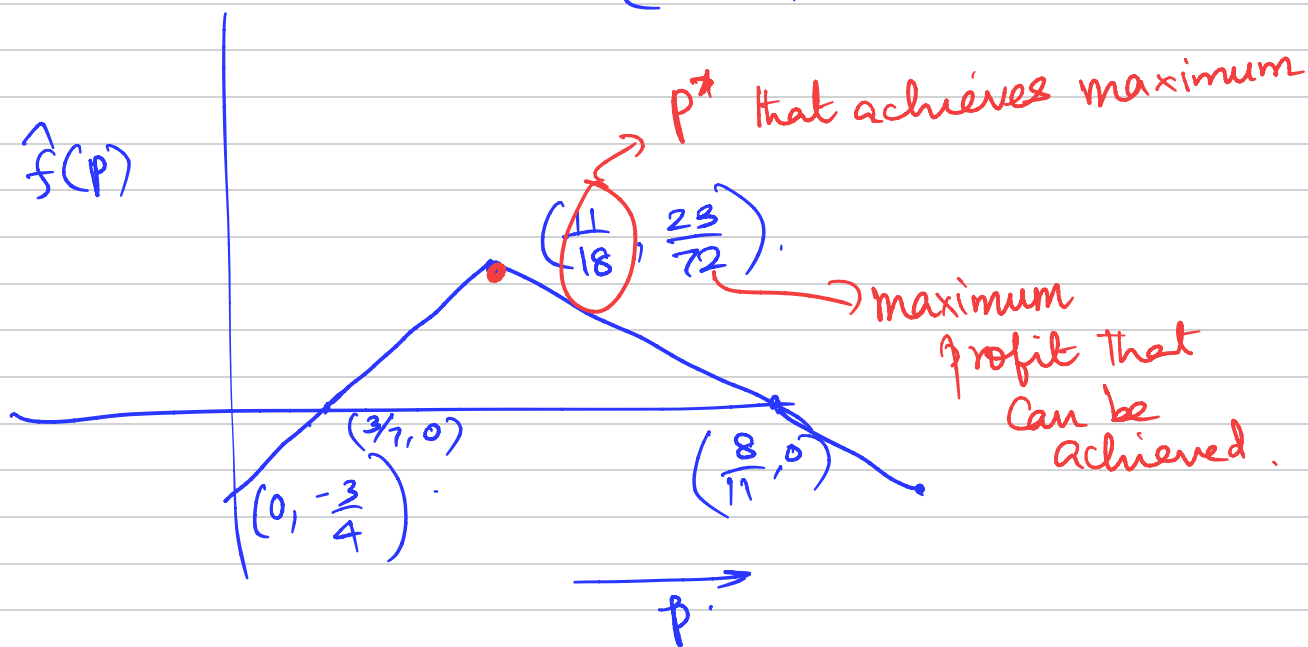
$$\max_p \left[ \min_k f(p, k) \right] = \hat{f}(p)$$

Note that  $k$  is the prob with which A will write 1, 2 respectively &  $p$  is the probability with which B will guess 1.

Analysis in previous page looks at

$$\hat{f}(p) = \min_k f(p, k)$$

$$\hat{f}(p) = \begin{cases} 2 - 11 \frac{p}{4} & p \geq \frac{11}{18} \\ \frac{7p-3}{4} & \text{otherwise} \end{cases}$$



$$\begin{aligned} & 2 - 11 \times \frac{11}{4 \times 18} \\ & \frac{8 \times 18 - 121}{4 \times 18} = \frac{23}{72} \\ & = \frac{7 \times 11 - 3}{4} \end{aligned}$$



$$\Rightarrow \arg \max_p \min \{E[X]\} = \frac{11}{18} \Rightarrow \max(\min(E[X])) = 2 - \frac{121}{72} = \frac{23}{72}$$

a) Now let  $X$ : units A gives to B = loss of A

$$P(X=x | B \text{ chooses } 1) = \begin{cases} q, & x=1 \\ 1-q, & x=-\frac{3}{4} \end{cases}$$

$$\Rightarrow E[X | B \text{ chooses } 1] = 2 - \frac{3}{4}(1-q) = \frac{7q-3}{4}$$

$$b) P(X=x | B \text{ chooses } 2) = \begin{cases} 1-q, & x=2 \\ q, & x=-\frac{3}{4} \end{cases}$$

$$E[X | B \text{ chooses } 2] = \cancel{2q - \frac{3}{4}(1-q)} \\ = (1-q)2 - \frac{3}{4}q = 2 - \frac{11q}{4}$$

$$E[X] = \left(\frac{7q-3}{4}\right)P(B \text{ chooses } 1) + \left(2 - \frac{11q}{4}\right)P(B \text{ chooses } 2) \\ = \left(\frac{7q-3}{4}\right)p + \left(2 - \frac{11q}{4}\right)(1-p) \\ = \left(\frac{18q-11}{4}\right)p + 2 - \frac{11q}{4}$$

Case I:  $18q-11 \geq 0, q \geq \frac{11}{18}$

$$\max E[X] = \frac{18q-11}{4} + 2 - \frac{11q}{4} = \frac{7q-3}{4}$$

$$\Rightarrow \arg \max_p \min_q E[X] = \frac{11}{18}$$

$$\min(\max E[X]) = \frac{23}{72}$$

Case II:  $18q-11 < 0$

$$\Rightarrow \max E[X] = 2 - \frac{11q}{4}$$

$$\Rightarrow \min(\max E[X]) = \frac{23}{72}$$

(Since  $E[X]$  here is less seen by A, A would like to minimize it given any strategy by B)



Q5 I ~~II~~ : no. of units ordered by customer from store

let  $X$  : no. of units sold,  $Z$  : profit

$$X|n = \begin{cases} I, & I \leq n \\ n, & I > n \end{cases}$$

$$\text{Profit} = b(\text{units sold}) + (-l)(\text{units unsold})$$

$$Z = bX - l(n - X)$$

$$\Rightarrow Z = (b + l)X - ln$$

$$E[Z] = \sum_{x=0}^{\infty} (b+l)E[X] - ln$$

$$= (b+l) \sum_{x=0}^{\infty} x P(x) - ln$$

$$= (b+l) \left[ \sum_{i=0}^n i P(i) + \sum_{i=n+1}^{\infty} n P(i) \right] - ln$$

$$= (b+l) \left[ \sum_{i=0}^n i P(i) + n \left( 1 - \sum_{i=0}^n P(i) \right) \right] - ln$$

$$= (b+l) \left[ \sum_{i=0}^n (i-n) P(i) \right] + bn$$

$$E[Z|n+1] = (b+l) \sum_{i=0}^{n+1} (i - \cancel{n}) P(i) + b(n+1)$$

$$= (b+l) \sum_{i=0}^{n+1} (i - (n+1)) P(i) + (b+l) \sum_{i=0}^{n+1} (-P(i)) + b(n+1)$$

$$= E[Z|n] + (b+l) \sum_{i=0}^n (-P(i)) + b$$

$$E[Z|n+1] - E[Z|n] = b - (b+l) \sum_{i=0}^n P(i) = -l_n$$

we have to increase  $n$  until  $e_n < 0$ .

$$e_n < 0 \Rightarrow (b+l) \sum_{i=0}^n p(i) < b$$

$$\Rightarrow \boxed{\sum_{i=0}^n p(i) < \frac{b}{b+l}}$$

Q9)

$$\begin{aligned} a) F_Y(y) &= P(Y \leq y) \\ &= 1 - P(Y > y) \\ &= 1 - P(\min(X_1, X_2, X_3) > y) \end{aligned}$$

$\min(X_1, X_2, X_3) > y \equiv$  each  $X_1, X_2, X_3 > y$   
as  $X_1, X_2, X_3$  are independent,

$$\begin{aligned} F_Y(y) &= 1 - P(X_1 > y) P(X_2 > y) P(X_3 > y) \\ &= 1 - (1 - F_{X_1}(y)) (1 - F_{X_1}(y)) (1 - F_{X_1}(y)) \\ &= 1 - (1 - F_{X_1}(y))^3 \end{aligned}$$

$$\begin{aligned} b) P(Y=y) &= F_Y(y) - F_Y(y-1) \\ &= 1 - (1 - F_{X_1}(y))^3 - \cancel{(1 - F_{X_1}(y-1))^3} \\ &\quad - (1 - (1 - F_{X_1}(y-1))^3) \\ &= (1 - F_{X_1}(y-1))^3 - (1 - F_{X_1}(y))^3 \end{aligned}$$

$$\begin{aligned} F_{X_1}(y) &= P(X \leq y) \\ &= \sum_{i=101}^y 0.1 = 0.1(y-100), \quad y \in \{101, 102, \dots, 110\} \end{aligned}$$

$$P(Y=y) = \begin{cases} (1 - F_{X_1}(y-1))^3 - (1 - F_{X_1}(y))^3, & y \in \{102, 103, \dots, 110\} \\ 1 - (1 - F_{X_1}(y))^3, & y = 101 \end{cases}$$



$$P_Y(y) = \begin{cases} \frac{(111-y)^3 - (110-y)^3}{1000}, & y \in \{102, 103, \dots, 110\} \\ 1 - (0.9)^3, & y = 101 \end{cases}$$

~~As~~ as  $P_{X1}(100) = 0$  anyway,

$$P_Y(y) = \frac{(111-y)^3 - (110-y)^3}{1000}, \quad y \in \{101, 102, \dots, 110\}$$

$$\begin{aligned} \text{ii) } E[Y] &= \sum_{y=101}^{110} y P_Y(y) \\ &= \sum_{y=101}^{110} \frac{y}{1000} [(111-y)^3 - (110-y)^3] \end{aligned}$$

$$= \frac{4121}{40} = 103.025$$

$$\begin{aligned} E[X] &= \sum_{x=101}^{110} x P_X(x) \\ &= \sum_{x=101}^{110} x (0.1) \\ &= \frac{211}{2} = 105.5 \end{aligned}$$

$\rightarrow$  expected score increased by 2.475

Q11)

Assume we play for  $n$  rounds, last round we got profit.

$$\begin{aligned} \text{Total profit (Net)} &= 10(-2^0 - 2^1 - 2^2 - \dots - 2^{n-2} + 2^{n-1}) \\ &= 10(- (2^{n-1} - 1) + 2^{n-1}) \\ &= 10 \end{aligned}$$



⇒ No matter how many rounds you play,  
net profit will always be 10.

$$P_X(x) = \begin{cases} 1, & x=10 \\ 0, & \text{otherwise} \end{cases}$$

where  $X$ : profit (net)

$$E[X] = 10(1) + 0 = 10$$

$X_i$ : profit at  $i^{\text{th}}$  round

$$E[X_i] = 0 \Rightarrow E\left[\sum_{i=1}^{\infty} X_i\right] = 0$$

But this is not ~~the~~ the expectation of NET profit,  
it takes out the fact that we stop when  
we get a win.

Q13)

$$a) P(Z=z) = P(X+Y=z)$$

$$= P(X=0, Y=z) + P(X=1, Y=z-1) \\ + \dots + P(X=z-1, Y=1) + P(X=z, Y=0)$$

$$= \sum_{k=0}^z P(X=k, Y=z-k)$$

independence ↗

$$= \sum_{k=0}^z P_X(k) P_Y(z-k)$$

$$= \sum_{k=0}^z \frac{e^{-\lambda_1} \lambda_1^k}{k!} \frac{e^{-\lambda_2} \lambda_2^{z-k}}{(z-k)!}$$

$$= e^{-(\lambda_1 + \lambda_2)} \lambda_2^z \sum_{k=0}^z \left(\frac{\lambda_1}{\lambda_2}\right)^k \binom{z}{k} \frac{1}{z!}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)} \lambda_2^z}{z!} \left(1 + \frac{\lambda_1}{\lambda_2}\right)^z$$

$$P_Z(z) = \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^z}{z!}$$

$$= \text{Poisson}(\lambda_1 + \lambda_2)$$

b)  $X$ : no. of 1's transmitted in a slot  ~~$\sim \text{Binom}$~~   
 $N$ : no. of packets in a slot

$$X|N=n \sim \text{Binom}(n, p)$$

$$\Rightarrow P(X=x|N=n) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \frac{P(X=x, N=n)}{P(N=n)}$$

$$\Rightarrow P(X=x, N=n) = \binom{n}{x} p^x (1-p)^{n-x} \left[ \frac{e^{-\lambda} \lambda^n}{n!} \right]$$

$$\leq \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x} \frac{e^{-\lambda} \lambda^n}{n!}$$

$$= \frac{e^{-\lambda} p^x (1-p)^{n-x}}{x!} \frac{e^{-\lambda} \lambda^n}{(n-x)!}$$

$$P(X=x) = \sum_{n=0}^{\infty} P(X=x, N=n)$$

$$= \frac{e^{-\lambda} p^x \lambda^n}{x!} \sum_{n=0}^{\infty} \frac{(\lambda(1-p))^{n-x}}{(n-x)!}$$

$$= \frac{e^{-\lambda} p^x \lambda^x}{x!} \sum_{m=0}^{\infty} \frac{(\lambda(1-p))^m}{m!}$$

$$= \frac{e^{-\lambda}}{x!} (p\lambda)^x e^{-\lambda(1-p)} e^{-\lambda p}$$



$$P_X(n) = \frac{(p\lambda)^n e^{-p\lambda}}{n!} \sim \text{Poisson}(p\lambda)$$

~~Q15~~ Q17)

a)  $S_i$  :  $i^{\text{th}}$  couple is alive (1) otherwise (0)

$$S = \sum_{i=1}^m S_i$$

⇒

$$E[S|A] = E\left[\sum_{i=1}^m S_i \mid A\right]$$

$$= \sum_{i=1}^m E[S_i \mid A]$$

$$= m E[S_1 \mid A]$$

↙ by symmetry



$$\begin{aligned}
 P(S_1 = 1 | A = a) &= \frac{P((S_1 = 1) \cap (A = a))}{P(A = a)} \\
 &= \frac{p^2 \left[ \binom{2m-2}{a-2} p^{a-2} p^{2m-a} \right]}{2^m \binom{a}{a} p^a p^{2m-a}} \\
 &= \frac{2^{m-2} \binom{a-2}{a-2}}{2^m \binom{a}{a}} \\
 &= \frac{a(a-1)}{2m(2m-1)}, \text{ Similarly } P(S_i = 1 | A = a) = \frac{a(a-1)}{2m(2m-1)}
 \end{aligned}$$

$$\Rightarrow E[S_1 | A] = \frac{a(a-1)}{2m(2m-1)}$$

$$E[S | A] = m \left[ \frac{a(a-1)}{2m(2m-1)} \right] = \frac{a(a-1)}{2(2m-1)}$$

b)  $E[S | A] = \sum_{s=1}^{\lfloor \frac{a}{2} \rfloor} s P(S | A)$

*there can only be upto  $\lfloor \frac{a}{2} \rfloor$  couples that have both members alive as total # of people alive are  $a$ .*

*possibilities of couples st only one person is alive*

$$\begin{aligned}
 &= \sum_{s=1}^{\lfloor \frac{a}{2} \rfloor} s \frac{\binom{m}{s} \left[ (p^2)^s \binom{m-s}{a-2s} 2^{a-2s} \right] p^{a-2s} (1-p)^{2m-a}}{\left[ 2^m \binom{a}{a} p^a (1-p)^{2m-a} \right]} \\
 &= \sum_{s=1}^{\lfloor \frac{a}{2} \rfloor} \frac{s}{2^m \binom{a}{a}} \binom{m-s}{a-2s} \binom{m}{s} (2^{a-2s}) \\
 &= \sum_{s=1}^{\lfloor \frac{a}{2} \rfloor} \frac{s}{(2^m) \binom{a}{a}} \frac{(m-s)! (2^{a-2s})}{(a-2s)! (m+s-a)!} \frac{m!}{(m-s)! s!} \\
 &= \sum_{s=1}^{\lfloor \frac{a}{2} \rfloor} \frac{m!}{2^m \binom{a}{a}} \frac{2^{a-2s}}{(s-1)! (a-2s)! (m+s-a)!}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \text{summation} &= \frac{E[S|A]}{m!} {}^{2m}C_a \\
 &= \frac{a(a-1)}{2(2m-1)} \frac{2m!}{a!(2m-a)!} \frac{1}{m!} \\
 &= \frac{1}{2(a-2)!} \frac{2m(2m-2)!}{(2m-a)!} \frac{1}{m!} \\
 &= \binom{2m-2}{a-2} \left( \frac{1}{(m-1)!} \right)
 \end{aligned}$$

Q19)

$$\begin{aligned}
 a) E[N_j | N_i > 0] P(N_i > 0) + E[N_j | N_i = 0] P(N_i = 0) \\
 = E[N_j]
 \end{aligned}$$

$$P(N_j = m) = {}^n C_m p_j^m (1-p_j)^{n-m} \sim \text{Binom}(n, p_j)$$

$$\Rightarrow E[N_j] = np_j$$

$$\text{similarly, } N_i \sim \text{Binom}(n, p_i)$$

$$\Rightarrow P(N_i = 0) = (1-p_i)^n$$

$$\begin{aligned}
 P(N_j = m | N_i = 0) &= \frac{P((N_j = m) \cap (N_i = 0))}{P(N_i = 0)} \\
 &= \frac{{}^n C_m p_j^m (1-p_j-p_i)^{n-m}}{(1-p_i)^n}
 \end{aligned}$$

$$\begin{aligned}
 E[N_j = m | N_i = 0] &= \frac{\sum_{m=0}^n m {}^n C_m p_j^m (1-p_j-p_i)^{n-m}}{(1-p_i)^n} \\
 &= \frac{np_j}{(1-p_i)^n} \sum_{m=1}^n {}^{n-1} C_{m-1} p_j^{m-1} (1-p_j-p_i)^{n-m} \\
 &= \frac{np_j}{(1-p_i)^n} (1-p_i)^{n-1} = \frac{np_j}{1-p_i}
 \end{aligned}$$



$$\Rightarrow E[N_j | N_i > 0] (1 - (1 - p_i)^n) + (1 - p_i)^n \frac{np_j}{(1 - p_i)^n} = np_j$$

$$E[N_j | N_i > 0] = np_j \left[ \frac{1 - (1 - p_i)^{n-1}}{1 - (1 - p_i)^n} \right]$$

$$b) P(N_j = m | N_i = 0) = \frac{{}^n C_m p_j^m (1 - p_j - p_i)^{n-m}}{(1 - p_i)^n}$$

$$\begin{aligned} P(N_j = m | N_i = 1) &= \frac{{}^n C_1 p_i^{n-1} {}^{n-1} C_m p_j^m (1 - p_i - p_j)^{n-m-1}}{{}^n C_1 (1 - p_i)^{n-1} p_i} \\ &= \frac{{}^{n-1} C_m p_j^m (1 - p_i - p_j)^{n-1-m}}{(1 - p_i)^{n-1}} \end{aligned}$$

$$E[N_j | N_i \leq 1] = \sum m P(N_j = m | N_i \leq 1)$$

$$\begin{aligned} P(N_j = m | N_i \leq 1) &= \frac{P((N_j = m) \cap ((N_i = 0) \cup (N_i = 1)))}{P((N_i = 0) \cup (N_i = 1))} \\ &= \frac{P((N_j = m) \cap (N_i = 0)) + P((N_j = m) \cap (N_i = 1))}{P(N_i = 0) + P(N_i = 1)} \end{aligned}$$

$$\begin{aligned} E[N_j | N_i \leq 1] &= \frac{\sum m P((N_j = m) \cap (N_i = 0)) + \sum m P((N_j = m) \cap (N_i = 1))}{P(N_i = 0) + P(N_i = 1)} \end{aligned}$$



$$E[N_j | N_i \leq 1] = \frac{n C_1 p_i (n-1) p_j (1-p_i)^{n-2} + n p_j (1-p_i)^{n-1}}{(1-p_i)^n + n(1-p_i)^{n-1} p_i}$$

(from part (a), summation is same)

$$E[N_j | N_i \leq 1] = \frac{n p_j (1-p_i)^{n-2} [(n-1) p_i + (1-p_i)]}{(1-p_i)^{n-1} [(1-p_i) + n p_i]}$$

$$P(N_i \leq 1) E[N_j | N_i \leq 1] = n p_j (1-p_i)^{n-2} [(n-2) p_i + 1]$$

$$\Rightarrow E[N_j | N_i > 1] [1 - (1-p_i)^n - n p_i (1-p_i)^{n-1}] + n p_j (1-p_i)^{n-2} ((n-2) p_i + 1) = n p_j$$

$$E[N_j | N_i > 1] = \frac{n p_j [1 - (1-p_i)^{n-1} - (n-1)(1-p_i)^{n-2}]}{1 - (1-p_i)^n - n p_i (1-p_i)^{n-1}}$$

(Q8)  $X_i: \begin{cases} X_i = 1 & \text{if } i^{\text{th}} \text{ coupon is used at least once (collected)} \\ X_i = 0 & \text{otherwise} \end{cases}$

$\Rightarrow X$ : no. of distinct coupons

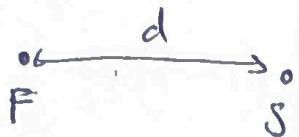
$$\rightarrow X = X_1 + X_2 + \dots + X_N$$

$$\Rightarrow E[X] = \sum_{i=1}^N E[X_i] \quad \left. \begin{array}{l} \text{symmetry} \\ \end{array} \right\} = N E[X_1]$$

$$P(X_1 = 1) = 1 - P(\text{1st coupon is never used}) = 1 - \frac{(N-1)^n}{N^n} \rightarrow \text{arbitrary selection with } N-1 \text{ options}$$

$$E[X] = N \left[ 1 - \left(1 - \frac{1}{N}\right)^n \right]$$

Q13)



let  $T_d$  : time taken when Fly and spider are  $d$  units apart, to land on same position

$t_d = E[T_d]$ ,  $X$  be  $\begin{cases} 1, & \text{fly moves right} \\ 0, & \text{fly stays of same pos} \\ -1, & \text{fly moves left} \end{cases}$   
 expected time for them to land in same position

By total law of expectation,

$$E[T_d] = E[E[T_d | X]]$$

$$= \sum_{x=-1}^1 P_X(x) E[T_d | X=x]$$

Initial :  $t_0 = 0$   
conditions

$$= P_X(-1) E[T_d | X=-1] + P_X(0) E[T_d | X=0] + P_X(1) E[T_d | X=1]$$

$$t_d = \underbrace{P t_d}_{\substack{\text{fly moves left} \\ \text{spider moves left} \\ \text{distance still } d}} + \underbrace{(1-2P) t_{d-1}}_{\substack{\text{fly : stay} \\ \text{spider : left} \\ \text{distance} = d-1}} + \underbrace{P t_{d-2}}_{\substack{\text{fly : right} \\ \text{spider : left} \\ \text{dist.} = d-2}}$$



## Correction

$$\text{let } t_d = E[T_d], \quad t_d = p(t_d + 1) + (1 - 2p)(1 + t_{d-1}) + p(1 + t_{d-2})$$

$$= 1 + p t_d + (1 - 2p) t_{d-1} + p t_{d-2}.$$

$$t_0 = 0.$$

$$t_1 = (1 - 2p)(1 + 0)$$

$$+ p(t_{-1} + 1)$$

$$+ p(t_1 + 1)$$

Spider  
doomed in this  
case where the

Spider crossed to left  
of the fly.

and will keep going left.

Fly can never catch up  
with spider.

It escapes !!

$$E[T_D] = \sum_{d \in \mathbb{D}} E[T_D | D=d] P(D=d)$$

$$= \sum_{d \in \mathbb{D}} t_d P(D=d)$$

$$X = 1 + I_{T_1}(1 + I_{H_2}(X' - 1)) + I_{H_1}(1 + I_{T_2}(X' - 1))$$

$\downarrow$  first toss       $\downarrow$  If first toss is tail       $\downarrow$  If first toss is heads

$$I_{T_1} = 1 - I_{H_1} = 1 \text{ if first toss is heads} \\ 0 \text{ otherwise}$$

$I_{T_1}$ ,  $I_{H_2}$  and  $X'$  are independent

$X$ ,  $X'$  follow same distribution

$$16. P(I=i, J=j) = P(I > \frac{n}{2}) P(I=i, J=j | I > \frac{n}{2}) + \\ P(I \leq \frac{n}{2}) P(I=i, J=j | I \leq \frac{n}{2})$$

$$P(I=i, J=j | I > \frac{n}{2}) = \frac{1}{\binom{n}{2}} \times \frac{1}{m_i} \text{ if } j \geq m - m_i \\ 0 \text{ otherwise}$$

$$P(I=i, J=j | I \leq \frac{n}{2}) = \frac{1}{\binom{n}{2}} \times \frac{1}{m_i} \text{ if } j \leq m_i \\ 0 \text{ otherwise}$$

Got the joint  $\rightarrow$  Can get marginal by summing over the other RV



$$E[\text{Marks}] = \sum_{i=1}^n v_i \left( \prod_{k=1}^i p_k \right) \rightarrow \text{If we see } i^{\text{th}} \text{ and } (i+1)^{\text{th}} \text{ term}$$

$$\rightarrow v_i p_1 p_2 \dots p_i + v_{i+1} p_1 p_2 \dots p_{i+1} = p_1 p_2 p_3 \dots p_{i-1} (v_i p_i + v_{i+1} p_i p_{i+1})$$

If the order of  $i^{\text{th}}$  and  $(i+1)^{\text{th}}$  term is reversed

then ~~we will get~~  $v_{i+1} p_{i+1} + v_i p_i p_{i+1}$  which is less than original

value we took assuming the inequality  $\rightarrow$

$$\text{for All } i, \frac{p_i v_i}{1 - p_i} > \frac{p_{i+1} v_{i+1}}{1 - p_{i+1}} \text{ for maximum expected marks}$$

2. At least one ball  $\geq 17 \rightarrow 1 - P(\text{All balls} < 17)$

$$\rightarrow p = 1 - \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} = \frac{61}{125}$$

4.  $E[X^n] = \sum_{x=0}^{\infty} x^n P(X=x)$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} x^n \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} \frac{(x+1)^n e^{-\lambda} \lambda^{x+1}}{(x+1)!}$$

$$= \sum_{x=0}^{\infty} \lambda (x+1)^{n-1} \frac{e^{-\lambda} \lambda^x}{(x)!}$$

$$E[X^n] = E[\lambda (X+1)^{n-1}] \rightarrow E[X^n] = \lambda E[(X+1)^{n-1}]$$

6.7) ~~Consider~~ Let probability of A correct is  $p_A$  and marks be  $V_A$ , similarly for B.

If A is chosen first then  $\rightarrow E[\text{Marks}] = V_A p_A + V_B p_A p_B$

If B is chosen first then  $\rightarrow E[\text{Marks}] = V_B p_B + V_A p_A p_B$

If ~~A~~ choosing A first gives more expectation of marks  $\rightarrow$

$$V_A p_A + V_B p_A p_B > V_B p_B + V_A p_A p_B \rightarrow \frac{V_A p_A}{1-p_A} > \frac{V_B p_B}{1-p_B}$$

If ~~A~~ choosing A gives more expectation then  $\rightarrow$

$$\frac{V_A p_A}{1-p_A} > \frac{V_B p_B}{1-p_B} \quad (\text{Similarly the other way})$$

If the choice of order for max expectation is 1, 2, ...

$$n \text{ then } \rightarrow E[\text{Marks}] = V_1 p_1 + V_2 p_1 p_2 + V_3 p_1 p_2 p_3 + \dots + V_n p_1 p_2 p_3 \dots p_n$$

~~the~~ the order is such  $\frac{p_i V_i}{1-p_i} > \frac{p_{i+1} V_{i+1}}{1-p_{i+1}}$



14) b)

Expected  
Tosses till you see two heads.

# tosses till you see two heads  $X = X_1 + X_2$

where  $X_1, X_2 \sim \text{Geometric}(p)$

$$\Rightarrow E[X] = E[X_1] + E[X_2] = \frac{2}{p}$$

(General version done in class, toss till you see  $r$  heads).

a) Toss till you see a heads.

$$X \sim \text{Geometric}(p) \Rightarrow E[X] = 1/p$$

c) Toss till you see <sup>two</sup> successive heads

$X$ : # tosses till you see two successive heads.

$H_1$ : event that first toss is head.

$T_1$ : " " " is tail

$$E[X] = p E[X|H_1] + (1-p) E[X|T_1]$$

$$E[X|T_1] = E[1+X] \quad (\text{first toss is tail so the problem resets}).$$

tosses are independent

$$E[X|H_1] = p(H_2|H_1) E[X|H_1H_2] + p(T_2|H_1) E[X|H_1T_2]$$

$$= p E[X|H_1H_2] + (1-p) E[X|H_1T_2]$$

$\parallel$   $2$                        $\parallel$   $2 + E[X]$

$$\Rightarrow E[X] = p [2 + (1-p)(2 + E[X])] + (1-p)(1 + E[X])$$

$$= p [2 + (1-p) E[X]] + (1-p) + (1-p) E[X]$$

check this calculation ↓

$$\Rightarrow p^2 E[X] = 1 + p \Rightarrow E[X] = \frac{1+p}{p^2}$$

④ Toss till you see two heads or tails.

$$E[X] = p E[X|H_1] + (1-p) E[X|T_1]$$

$$E[X|H_1] = p \underbrace{E[X|HH_2]}_{2''} + (1-p) \underbrace{E[X|HT_2]}_{\boxed{HT \dots}}$$

$$\Rightarrow E[X|HT_2] = E[1+X|T_1]$$

$$\Rightarrow \underbrace{E[X|H_1]}_a = 2p + (1-p) \left[ 1 + \underbrace{E[X|T_1]}_b \right] \rightarrow \textcircled{1}$$

Similarly

$$E[X|T_1] = p \underbrace{E[X|T_1H_2]}_{1 + E[X|H_1]} + (1-p) \underbrace{E[X|T_1T_2]}_{2''}$$

$$\Rightarrow E[X|T_1] = p(1 + E[X|H_1]) + 2(1-p) \rightarrow \textcircled{2}$$

Solve for eqns  $\textcircled{1}, \textcircled{2}$ .

please check.

$$\textcircled{2} \text{ in } \textcircled{1} \Rightarrow a = 2p + (1-p) [1 + p(1+a) + 2(1-p)]$$

$$\Rightarrow a [1 - p(1-p)] = 2p + (1-p)(1+p+2-2p)$$

$$\rightarrow a = \frac{2p + (1-p)(3-p)}{1-p+p^2}$$

Similarly solve for b.

$$E[X] = pa + (1-p)b.$$

(18)

$$E[X_1 + X_2 + \dots + X_N] = \sum_{n=1}^{\infty} P(N=n) E[X_1 + X_2 + \dots + X_n | N=n]$$

total law of expectation

$$= \sum_{n=1}^{\infty} P(N=n) E[X_1 + X_2 + \dots + X_n | N=n]$$

linearity of

$$= \sum_{n=1}^{\infty} P(N=n) \sum_{i=1}^n E[X_i | N=n]$$



expectation

$$\begin{aligned}
 &= \sum_{n=1}^{\infty} P(N=n) \left( \sum_{i=1}^n E[X_i] \right) \\
 &= \sum_{n=1}^{\infty} P(N=n) n E[X_1] \\
 &= E[X_1] E[N]
 \end{aligned}$$

*X<sub>i</sub>'s are independent of N.*

*X<sub>i</sub>'s are identical*

$$\begin{aligned}
 10) \sum_{i=0}^{\infty} i P(X=i) &= \sum_{i=1}^{\infty} i P(X \geq i) = \sum_{i=1}^{\infty} i \sum_{j=i}^{\infty} P_X(j) \\
 &= \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} i P_X(j) \\
 &= \sum_{j=1}^{\infty} \sum_{i=1}^{j-1} i P_X(j) \\
 &= \sum_{j=1}^{\infty} \frac{(j-1)j}{2} P_X(j) \\
 &= \frac{E[X^2] - E[X]}{2}
 \end{aligned}$$

⑫  $E[X | X+Y=m]$        $X, Y \sim \text{Binomial}(n, p)$   
 $X, Y$  independent.

$$P_{X|\{X+Y=m\}}(x) = \frac{P(\{X=x\} \cap \{X+Y=m\})}{P(\{X+Y=m\})}$$

$$X+Y \sim \text{Binomial}(2n, p)$$

$$= \frac{P(\{X=x\} \cap \{Y=m-x\})}{P(X+Y=m)}$$

$$= \frac{\binom{n}{x} \cancel{p^x} \cancel{(1-p)^{n-x}} \binom{n}{m-x} \cancel{p^{m-x}} \cancel{(1-p)^{n-m+x}}}{P(X+Y=m)}$$

$$\begin{cases} \binom{2n}{m} p^m (1-p)^{2n-m} & x \leq m \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{\frac{n!}{x!(n-x)!} \frac{n!}{(m-x)!(n-m+x)!}}{\frac{2n!}{m!(2n-m)!}} & x \leq m \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{\left[ \frac{n!}{x!(n-x)!} \right] \left[ \frac{(2n-m)!}{(n-x)!(n-m+x)!} \right]}{\left( \frac{2n!}{n!n!} \right)} & x \leq m \\ 0 & \text{otherwise} \end{cases}$$

Gives us two  
 combinatorial identities  
 $\sum_{x=0}^m \binom{m}{x} \binom{2n-m}{n-x} = \binom{2n}{n}$   
 $x=0$  and  $\sum_{x=0}^m \binom{n}{x} \binom{n}{m-x} = \binom{2n}{m}$

$$= \begin{cases} \frac{\binom{m}{x} \binom{2n-m}{n-x}}{\binom{2n}{n}} & x \leq m \\ 0 & \text{otherwise} \end{cases}$$

$$E[X | X+Y=m] = \sum_{x=0}^m x \frac{\binom{m}{x} \binom{2n-m}{n-x}}{\binom{2n}{n}}$$

$$x \binom{n}{x} = n \binom{n-1}{x-1}$$

$$\text{let } \hat{x} = x-1$$

$$\begin{aligned}
 &= \sum_{x=0}^m x \frac{\binom{m}{x} \binom{n}{m-x}}{\binom{2n}{m}} \\
 &= \sum_{x=1}^m n \frac{\binom{n-1}{x-1} \binom{n}{m-x}}{\binom{2n}{m}}
 \end{aligned}$$

$$= \sum_{\hat{x}=0}^{m-1} n \frac{\binom{n-1}{\hat{x}} \binom{n}{m-1-\hat{x}}}{\binom{2n}{m}}$$

Claim:

$$\sum_{\hat{x}=0}^{m-1} \binom{n-1}{\hat{x}} \binom{n}{m-1-\hat{x}} = \binom{2n-1}{m-1}$$

$\swarrow$  # of ways to pick  $\hat{x}$  elements from first  $n-1$  elements  
 $\swarrow$  # of ways to pick  $m-1-\hat{x}$  elements from last  $n$  elements  
 $\swarrow$  # of ways to pick  $m-1$  elements from a set of size  $2n-1$

$$\Rightarrow E[X | X+Y=m] = \frac{n \times \binom{2n-1}{m-1}}{\binom{2n}{m}} = \frac{n \times m}{2n} = \frac{m}{2}$$

Intuitively this is expected as  $X$  and  $Y$  are identical given  $X+Y=m$ , Expectation of  $X$  needs to be  $m/2$ .

Simpler proof:

$$E[X | X+Y=m] + E[Y | X+Y=m] = E[X+Y | X+Y=m] = m$$

Since  $X, Y$  are identical; independent

$$E[X | X+Y=m] = E[Y | X+Y=m] = u$$

$$\Rightarrow 2u = m \Rightarrow u = m/2$$